

SURAT, India

WMO: 428400

Lat: 21.200N

Lon: 72.833E

Elev: 12

StdP: 101.18

Time Zone: 5.50 (IND)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	14.1	15.4	3.3	4.8	25.4	5.1	5.4	24.8	4.8	24.1	4.0	24.5	0.6	0	0.324

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
5	6.4	38.2	22.6	36.7	22.9	35.5	23.3	28.4	32.1	28.1	31.7	27.7	31.2	1.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.5	23.4	30.3	27.2	22.9	30.0	26.9	22.5	29.8	92.0	32.1	90.5	31.8	89.0	31.4	29.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
5.3	4.4	3.6	11.9	40.8	1.6	1.4	10.7	41.8	9.7	42.6	8.8	43.4	7.6	44.3		
			DB	WB	9.1	29.1	1.6	0.5	7.9	29.5	7.0	29.7	6.1	30.0	5.0	30.3

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	27.8	22.8	24.8	28.3	30.4	31.3	30.4	28.6	28.2	28.6	29.1	27.0	24.0
	DBStd	3.05	1.91	2.36	2.09	1.57	1.28	1.71	1.31	1.03	1.23	1.49	1.67	1.88
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	1	0	1	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6499	397	416	568	613	661	612	575	563	558	592	509	436
	CDD18.3	3458	138	183	309	363	403	362	317	305	308	334	259	177
	CDH23.3	40633	1062	1654	3575	4823	5634	4939	3801	3463	3618	4061	2604	1396
Wind	WSAvg	1.3	1.1	1.0	1.0	1.3	2.1	2.2	2.0	1.7	1.0	0.6	0.7	0.9
	PrecAvg	1252	0	0	1	0	3	258	447	298	175	20	17	2
	PrecMax	2299	1	9	35	4	25	837	884	830	399	148	174	37
	PrecMin	564	0	0	0	0	0	1	48	37	15	0	0	0
	PrecStd	425	0	2	7	1	7	200	212	203	113	39	38	8
	DB	32.9	35.9	39.1	40.2	39.8	36.1	33.3	32.2	34.7	36.7	35.2	33.2	33.2
	MCWB	20.0	20.4	21.3	21.0	23.9	27.2	27.2	26.6	26.6	24.1	22.2	20.6	20.6
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	2% DB	31.1	34.1	37.2	38.4	37.3	34.7	32.0	31.2	33.1	35.6	34.0	31.9	31.9
	2% MCWB	19.2	20.2	20.7	22.2	24.8	27.3	27.2	26.5	26.5	23.9	21.8	20.3	20.3
	5% DB	29.8	32.6	35.7	37.0	35.4	33.9	31.2	30.6	32.1	34.7	33.1	30.8	30.8
	5% MCWB	18.9	19.9	20.7	22.7	25.8	27.3	27.1	26.4	26.4	23.7	21.8	20.0	20.0
	10% DB	28.5	31.0	34.4	35.4	34.0	33.2	30.6	30.0	31.2	33.7	32.2	29.6	29.6
	10% MCWB	18.4	19.4	20.6	22.8	26.5	27.3	27.0	26.3	26.1	23.9	21.7	19.5	19.5
	WB	22.0	23.3	25.8	27.6	28.7	28.8	28.4	27.9	28.2	28.3	26.6	23.7	23.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	MCDB	28.9	30.3	30.7	32.6	32.4	32.7	31.3	30.3	31.5	32.3	31.4	30.0	30.0
	2% WB	21.0	22.3	24.7	26.9	28.2	28.5	27.8	27.2	27.7	27.6	25.7	22.5	22.5
	2% MCDB	28.1	28.9	29.6	31.4	32.1	32.3	30.5	29.6	30.7	31.4	29.9	28.9	28.9
	5% WB	20.3	21.6	23.9	26.5	27.9	28.1	27.6	27.0	27.3	27.1	24.8	21.5	21.5
	5% MCDB	27.2	28.4	28.8	30.8	31.8	31.9	30.1	29.3	30.1	30.8	29.0	28.0	28.0
	10% WB	19.5	20.8	23.1	26.0	27.6	27.8	27.2	26.7	26.9	26.6	23.7	20.6	20.6
	10% MCDB	26.4	27.7	28.5	30.3	31.5	31.5	29.7	28.9	29.7	30.3	28.6	27.3	27.3
Mean Daily Temperature Range	MDBR	12.5	12.4	12.1	9.9	6.4	4.6	3.4	3.4	5.0	9.7	11.5	12.2	12.2
	5% DB	13.9	14.4	14.2	13.0	9.1	5.5	4.3	4.2	6.5	11.9	12.8	13.5	13.5
	5% WB	5.4	5.0	4.6	4.2	3.2	1.8	1.8	1.6	2.3	3.9	4.5	5.2	5.2
	MCDBR	12.1	12.1	11.2	8.7	6.3	5.0	3.9	3.7	5.7	7.6	9.2	12.0	12.0
	MCWBR	5.6	5.1	4.2	3.4	2.1	2.0	2.0	1.8	2.5	3.3	4.2	5.5	5.5
Clear-Sky Solar Irradiance	taub	0.449	0.443	0.491	0.537	0.572	0.686	0.701	0.644	0.595	0.565	0.539	0.495	0.495
	taud	2.116	2.148	2.029	1.961	1.927	1.717	1.695	1.813	1.911	1.920	1.934	1.989	1.989
	Ebn at Noon	812	847	821	787	753	666	657	697	726	726	718	747	747
	Edh at Noon	145	149	174	189	194	237	243	217	194	185	173	160	160
All-Sky Solar Radiation	RadAvg	4.77	5.65	6.59	7.20	7.27	5.49	3.90	3.99	4.82	5.42	4.78	4.47	4.47
	RadStd	0.14	0.16	0.18	0.13	0.23	0.45	0.44	0.37	0.42	0.27	0.22	0.21	0.21

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page